



► Cyber Security in Hospitals

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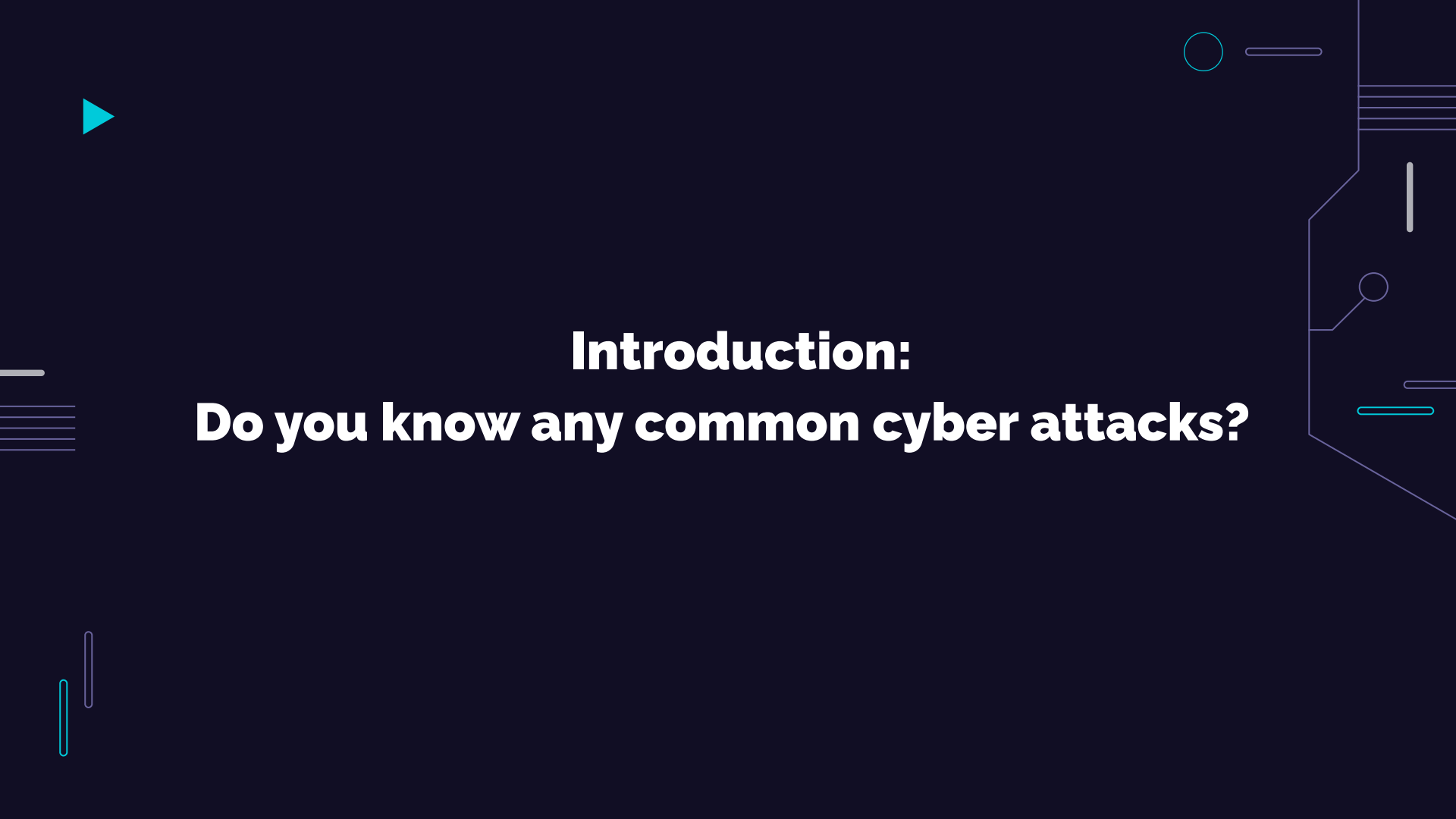
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Introduction:

Do you know any common cyber attacks?

► Prompt

YWCC As the digital world continues to grow, so do the volumes, variety, and velocity of cyber threats and attacks. The world is awash in data, and there is always someone trying to turn it into their own virtual currency. Today malware and ransomware are hitting everything from our personal cell phones to mission-critical infrastructure and supply chains. Your team has been hired by a large hospital system to do an analysis of digital security threat that it may be facing. You are asked to present the current state of threats, various modes of attacks, and how should the health system prepare itself.

► Introduction

- Cyber crime has become a persistent threat to many industries, with many attacks being undetected or unreported.
 - From 2015, were global cyber crime costs around 3 trillion, 2021 jumped to over 6 trillion
- The range of a cybercrime can be committed by lone people / smaller groups and can be as large as nation-state actors
 - Financial issues
 - Destroying or compromising victim data
 - Politics and national terrorist movements



Hospitals

Issues

- Bovid = Wake up call
- Healthcare faces larger cyber risks
 - Most targeted sectors globally
 - 81% of 223 organizations surveyed
 - >110 million patients attacked in 2015
 - 2018 - 2023
 - 278% ransomware attacks
 - 93% increase in data breaches



► The Reason

Heavy Intake of Information

- To do their jobs, healthcare professionals need to collect and analyze the most intimate details of our personal lives
 - Managed correctly, especially given the new technology now available, collection of that information can contribute greatly to the recovery of a patient.
- But without the correct protocols in place, that very same information can lead to inadvertent harm, possibly in excess of the original reason for seeking medical attention.



► A Security Systems Goals

The issue is that cyber security issues shouldn't be a heavy focus on the hospitals part, but on the patients part. In depth data collected by hospitals can help save patients lives, as long as that information is handled correctly. Cyber security can be a patient's saving grace or ultimate harm depending on how it is handled with the hospital.

Integral Aspects:

1. Maintain confidentiality: meaning that access to information is restricted only to authorized users
2. Maintain integrity: meaning the information is trustworthy and accurate
3. Maintain availability: meaning authorized users can quickly and reliably access information

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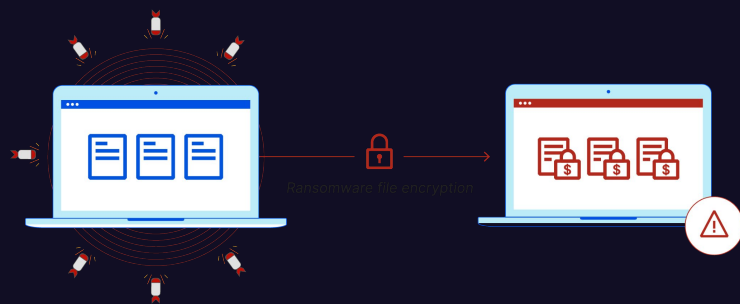
Common Cyber Attacks

► COMMON CYBER ATTACKS

1. Ransomware & Extortion - this is when hackers encrypt data and threatens public release unless paid.
2. Phishing and Social Engineering - Deceptive emails or calls to steal credentials or deliver malware
3. DDoS Attacks - Overwhelms network availability, halt clinical services, crashing the infrastructure.

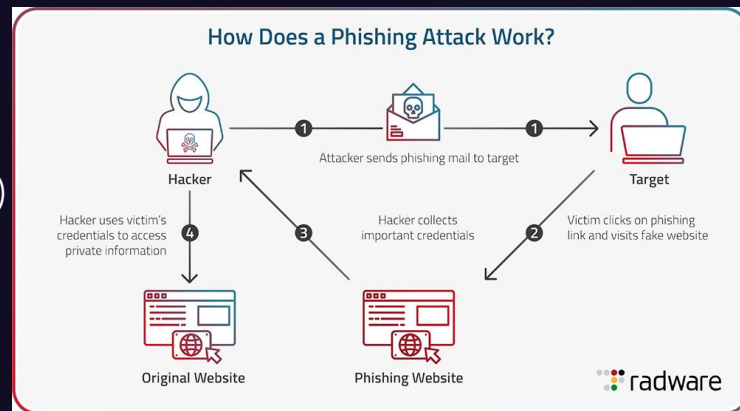
► Common Cyber Attacks: Ransomware

1. Scale of Incidents: Over 630 global healthcare ransomware incidents reported in 2023. Of which 460+ were affected. (I.S. Partners)
2. Rapid Growth: U.S. Healthcare Sector victims increased up 128% in 2023. (I.S. Partners)
3. How Ransomware Works:
 - a. Ransomware encrypts files and systems, denying access until a ransom is paid for the decryption key
 - b. Attackers utilize phishing emails, exploit kits, or unpatched vulnerabilities to gain entry and deploy encryption tools
 - c. Finally they issue extortion demands often exfiltrating data first for leverage.



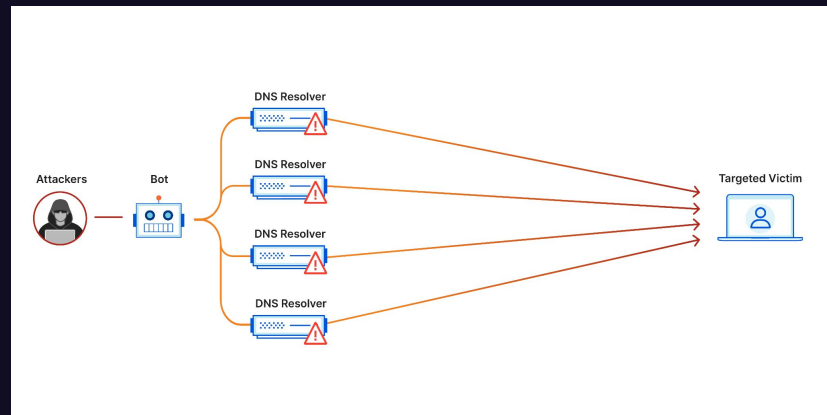
► Common Cyber Attacks: Phishing and Social Engineering

1. Over 90% of healthcare cyber attacks in 2024 began with phishing (Varonis)
2. 88% of healthcare employees clicked on simulated phishing emails during security drills in 2024. (Varonis)
3. How does it work?
 - a. Phishing is where attackers impersonate trusted entities in emails, texts, or calls
 - b. Messages contain malicious links or attachments that when opened, harvest login data or drop viruses
 - c. Attackers use phishing to reveal credentials or download malware



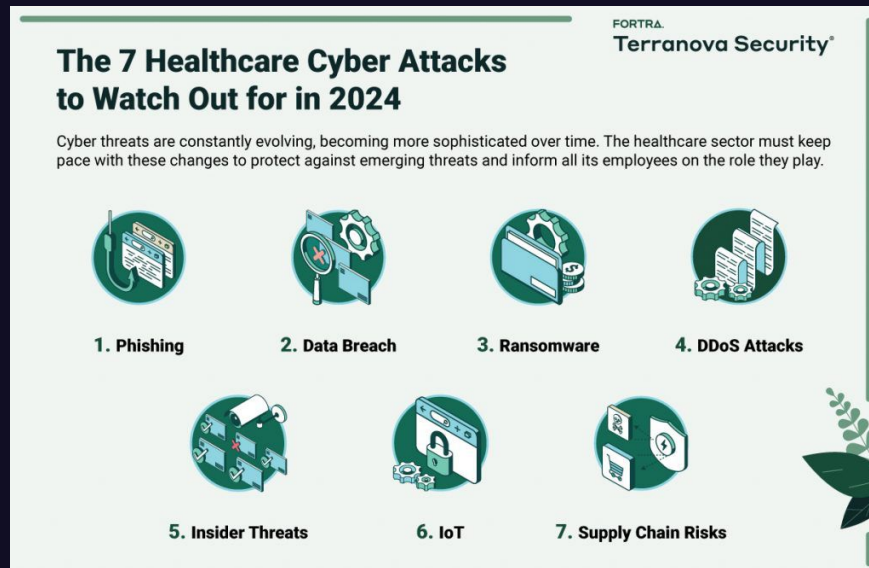
► Common Cyber Attacks: Distributed Denial of Service (DDoS)

1. DDoS attacks on critical healthcare infrastructure climbed by 55% over the last four years (Techtarget)
2. In 2024, Layer 7 DDoS attacks against North American Pharmaceutical firms outpace the total volume seen in 2023 (Techtarget)
3. How it works?
 - a. A DDoS attack overwhelms a network or server with excessive traffic from multiple compromised devices overwhelming resource limits and preventing access
 - b. Attacks may launch floods, protocol exploits, or application-layer requests
 - c. These outages can interrupt electronic health records, telemedicine platforms, and critical monitoring systems.



► Common Cyber Attacks: Impacts on Healthcare Operations

1. **Patient Safety Risks:** 69% of Healthcare organizations report care disruptions and medical complications during cyber incidents (I.S. Partners)
2. **Data Exposure Crisis:** 168 million healthcare records exposed in 2023.
3. **Severe Financial Burden:**
 - a. On Average data breach cost in healthcare: \$9.77 million.
 - b. Change healthcare attack caused up to \$1 billion daily loss
4. **Impact Treatment:** DDoS can cause many healthcare companies to not help patients delaying treatment or causing death.



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Common Malware Solutions

► What are some common malware solutions?



AWARENESS

Human error is behind over 90% of cyberattacks. Teaching users how attacks look in real life drastically lowers risk.



RECOVERY

If data is encrypted or held hostage, an up-to-date backup means you can restore systems quickly — no need to pay the ransom.



PREVENTION

Implementing proactive measures, policies, and controls designed to minimize security vulnerabilities and inhibit the success of malicious attacks.

► Solutions to Ransomware & Double Extortion

- Regular, Immutable Backups: Offline or cloud backups that can't be encrypted by attackers.
- Zero Trust Architecture: Restrict lateral movement; verify every access attempt.
- Endpoint Detection & Response (EDR): Detects malicious behavior before encryption completes.
- User Training: Recognize suspicious links/files; avoid opening unknown attachments.
- Incident Response Plan: Immediate isolation, assessment, and recovery process.

► Solutions to Phishing & Social Engineering

- Multi-Factor Authentication (MFA): Even if credentials are stolen, accounts stay protected.
- Security Awareness Training: Teach users how to identify phishing emails, spoofed links, suspicious requests.
- Email Filtering & Threat Intelligence: Block known malicious IPs/domains and scan for suspicious content.
- Simulated Phishing Tests: Gauge employee readiness and improve resilience.

► Solutions to DDoS Attacks

- Traffic Filtering & Rate Limiting: Drop malicious packets and throttle suspicious requests.
- CDNs & Cloud DDoS Protection: Absorb and mitigate large-scale attacks (e.g., Cloudflare, Akamai).
- Redundant Infrastructure: Load balancing, geo-distributed servers.
- Incident Playbooks & Monitoring: Alert thresholds and predefined responses to mitigate downtime.

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Proposal For Healthcare Companies

► Prevention

Security Policies and Training

- Clear, easy-to-follow guidelines for all staff
- Annual refresher on phishing and safe device use
- Remind all staff to report anything they suspect

Basic Technical Hygiene

- Monthly patching schedule for servers and workstations
- Inventory of all connected devices
- Updating and securing all devices
- Installation of company wide malware software

Access Controls

- Enforcing strong passwords and simple multi-factor authentication (DUO)
- Limiting administrative rights to only those who need them

► Detection and Response

Monitoring and Alerts

- Enables basic log collection on firewalls and key servers
- Configure email notifications for unusual login attempts

Incident Response Plan

- Create a checklist: who to call, steps to isolate affected systems, when to notify leadership
- Hold a Annual tabletop drill with IT and department heads

Clear Reporting Channels

- Designate a simple point of contact such as an IT department
- Create a simple form or hotline for staff to report suspicious emails or behavior

► Recovery and Continuous Improvement

Regular Backups and Restore Tests

- Do weekly encrypted, offline backups of patient data and key systems
- Do quarterly test restores to verify backups

Disaster Recovery Basics

- Document essential systems and acceptable downtime
- Identify a fallback location

Review and Updates

- Hold a Post-incident review to capture lessons learned
- Annual review of policies, inventory, and contacts



**Thank
You!**
Q & A

► Resources

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